



Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322: Computer Peripheral and Interfacing Sessional

Project Report



Project Title: TrashBot

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1 Introduction

With the swift development of Robotics, this problem of public cleanliness can be solved. In some highly populated areas, waste collection is done manually, which can be a time-consuming and inefficient process. To overcome this we introduce TRASHBOT, an autonomous IoT (Internet of Things) robot for the easy collection of waste in public areas. TRASHBOT combines the autonomous navigation system with the robotic arm that is responsible for detecting, collecting and segregating the waste. The operational modes include Autonomous Mode (the system autonomously navigates around obstacles to reach a destination) and Manual Mode (the system is controlled by a person from afar using an Android app). The mechanical automation and real-time IoT data monitoring make TRASHBOT an effective, sustainable, and affordable way to keep cities cleaner.

2 Project Objectives

The primary goal of this project is to develop an intelligent and affordable robotic system that can assist in public waste management while reducing dependence on manual labor.

The specific objectives of the project are as follows:

1. To develop an autonomous navigation mechanism that enables the robot to move independently, detect obstacles, and follow a safe path using ultrasonic sensors.
2. To design and implement a robotic arm capable of picking up waste materials from the ground and placing them into designated storage compartments.
3. To incorporate an automatic waste segregation feature that can distinguish between metallic and non-metallic objects using suitable sensing technology.
4. To integrate IoT-based communication for real-time monitoring and remote control through a mobile application.
5. To ensure reliable operation of the system by implementing an efficient power management solution using rechargeable batteries and voltage regulation circuits.

3 Problem Statement

The TRASHBOT system will solve the following problems that are important in the present waste management system of urban areas: The increase of uncollected refuse in high density public places, like railway stations, bus terminals, pollutes the environment, becomes breeding ground for pathogens and causes severe sanitary risks. Conventional Manual Sanitation is labour intensive, expensive and not efficient enough to cover high quantities at a large scale. Moreover, it puts sanitation workers at risk of exposure to

dangerous (unhygienic) conditions and toxic substances. The current automated cleaning technologies are mostly limited to indoor conditions with a structured layout. They are not autonomous and adaptable enough to work in the unpredictable environments outside and have no inbuilt telemetry (IoT) for immediate fleet management. Other problems are the reliance on human effort to ensure that sanitation standards are consistently met 24x7, especially when dealing with large public infrastructure which is simply not feasible logistically and/or financially to monitor all the time.

4 Scope

TRASHBOT project mainly studies the design and prototyping of a semi-autonomous rover for **Dry Waste Management in public areas**. The minimal set of functions covers:

1. **Dual-Mode Operation:** Autonomous Obstacle Avoidance and Manual Teleoperation using an IoT enabled Android app.
2. **Automated Collection & Segregation:** Designing an automated system of a robotic arm for physical collection of waste materials and a waste material segregation mechanism based on inductive sensing.
3. **Remote Telemetry:** Connection of live video transmission and system status to remote operators.
4. **Prototype Limitations:** Being optimized for flat terrain and solid dry waste, detection by the sensor is around 30 cm.

5 Methodology

The TRASHBOT project is based on an agile prototyping approach, where the hardware of the robot is divided into three subsystems: Locomotion, Manipulation and IoT Communication.

5.1 System Architecture

It is based on a centralized control model using the **Raspberry Pi 3 Model B**. The Raspberry Pi runs Python control logic, controls video streaming and sends telemetry to the Blynk cloud using the built-in Wi-Fi. It processes data from ultrasonic sensors & proximity sensors and drives DC motors (Navigation) and Servo motors (Waste collection). 11.1V Li-Ion battery with buck converters provide stable power distribution to all components.

5.2 Technology Stack

Key hardware and software components are listed in Table 1.

Table 1: Technology Stack of TRASHBOT

Category	Component / Technology
Controller	Raspberry Pi 3 Model B (Quad Core 1.2GHz)
OS & Logic	Raspbian OS / Python
Connectivity	Wireless Internet (802.11n) Built-in
Sensors	HC-SR04 (Obstacle), Inductive Proximity (Metal)
Actuation	MG995 Servos & L298N Motor Driver
Power	11.1V 10000mAh Li-Ion, Buck Converters
Interface	Blynk IoT App, IP Webcam
Chassis	4-Wheel Differential Drive Platform

6 Budget Estimation

The following is a detailed breakdown of the estimated costs of the hardware components needed for the TRASHBOT project (Table 2).

Category	Component Name	Unit Price (BDT)	Total (BDT)
Controllers	Raspberry Pi 3 Model B (1GB)	10,199	10,199
	Raspberry Pi AI Camera	6,880	6,880
	ESP-32 Development Board	550	550
Motors & Drivers	Gripper	680	680
	Tower Pro MG995 servo motor	354	708
	Gear Motor with Wheel	155	620
	Motor Driver	890	890
Sensors	Ultrasonic Sonar Sensor	110	660

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Category	Component Name	Unit Price (BDT)	Total (BDT)
	Metal Proximity Switch Sensor	390	390
	IR LED Sensor	150	150
	LED Sensor	99	99
	Alcohol Gas Sensor	144	144
	Dual IR Flame Sensor	450	450
	Water Sensor	350	350
	Vibration Sensor	380	380
Power	Transformer 12V 3A	390	390
	Buzzer	15	15
Mechanical & Misc	Full-Size Bread-board	120	240
	Half-Size Bread-board	75	150
	Jumper Wires (Male-Male)	100	400
	Li-ion Battery	260	1,040
	Solar Panel	350	350
	Battery case	50	50
	Display	340	340
Module	GSM GPRS Module	1,560	1,560
	GPS Module	440	440
Grand Total Estimated Cost			28,125 BDT

Table 2: Detailed Budget Estimation

7 Project Structure

7.1 Circuit Diagram

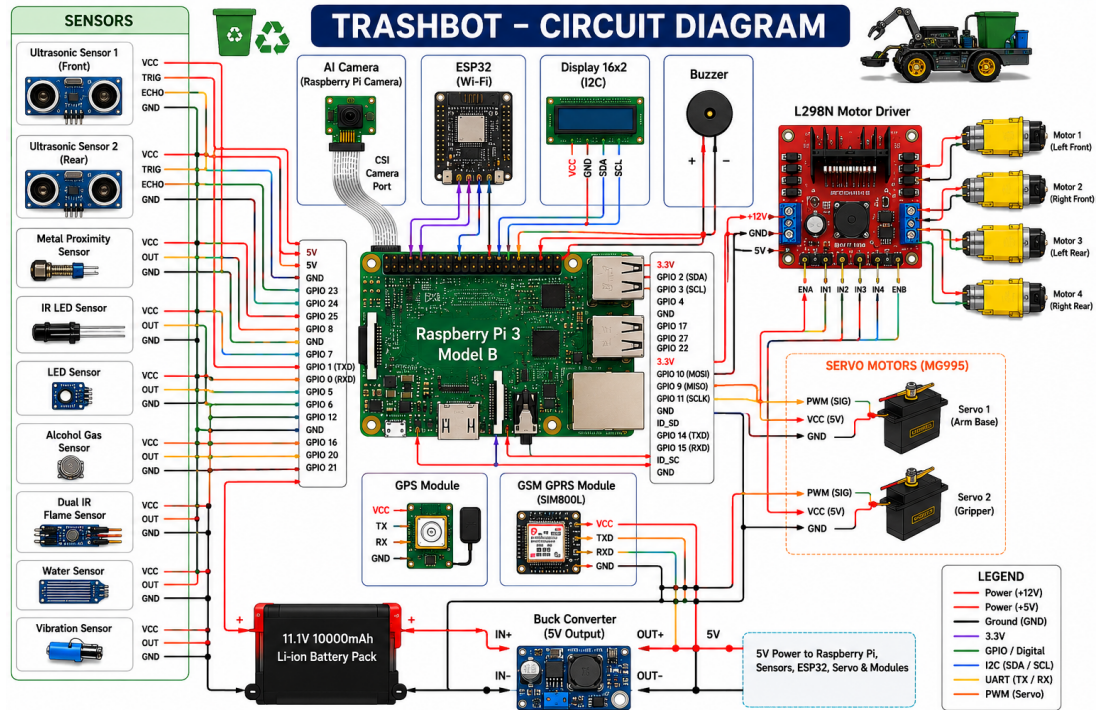


Figure 1: Project Circuit Diagram

7.2 Flowchart

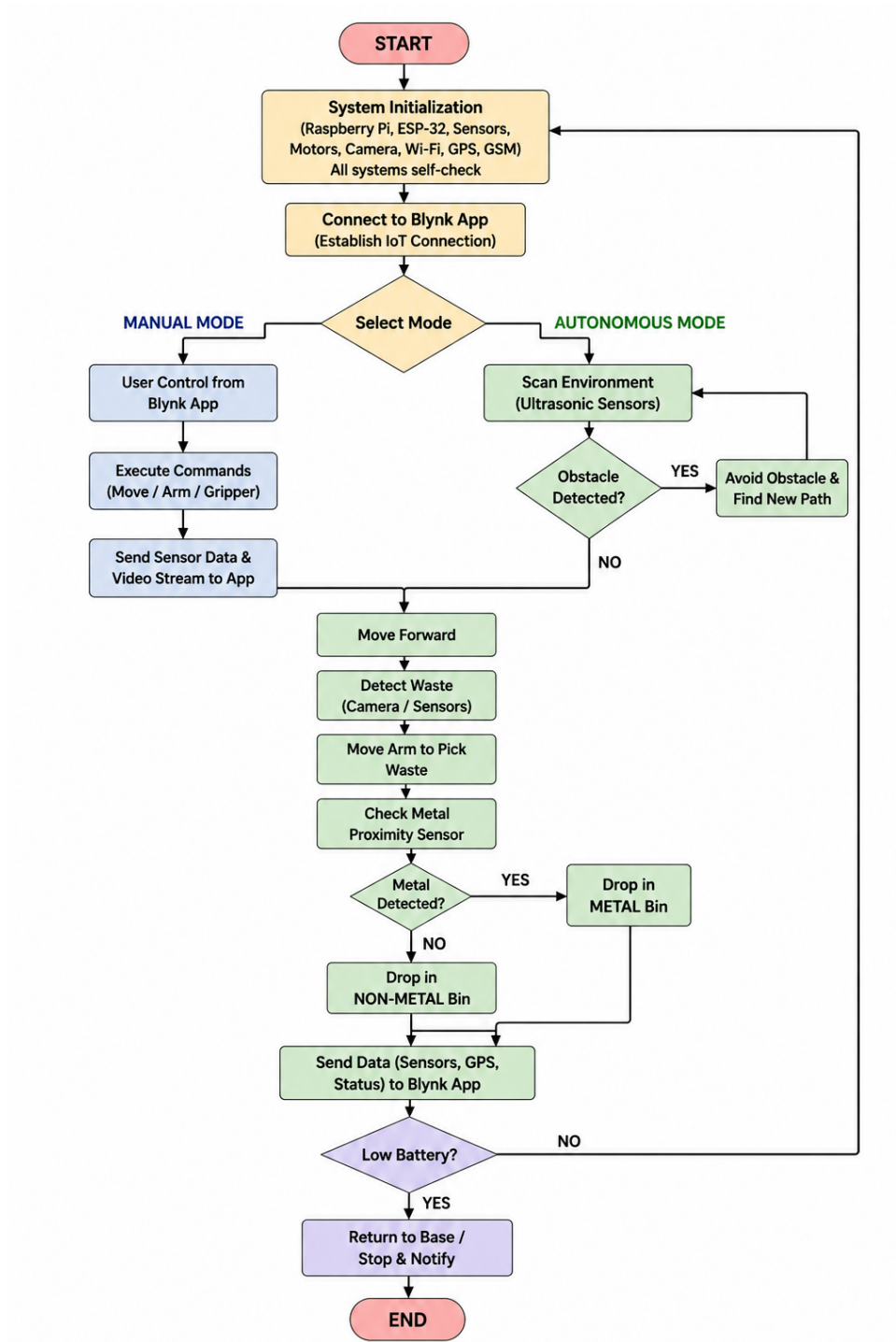


Figure 2: Project Flowchart

8 Implementation

The TRASHBOT system was implemented using a Raspberry Pi 5 as the central controller. Ultrasonic sensors were used for obstacle detection and autonomous navigation,

while DC motors enabled robot movement. A robotic arm driven by servo motors was developed to collect waste materials. A metal proximity sensor was integrated for automatic waste segregation into metallic and non-metallic categories. An ESP32-CAM provided live video monitoring, and IoT communication enabled remote control and status monitoring through a mobile application. The system was powered by rechargeable Li-ion batteries with voltage regulation circuits to ensure stable operation.

9 Gantt Chart

Table 3: Project Timeline

Task	W1	W2	W3	W4	W5	W6
Project planning and requirement analysis	✓					
Component selection and circuit design		✓				
Raspberry Pi and sensor integration		✓	✓			
Robotic arm development			✓	✓		
Waste segregation implementation				✓	✓	
IoT communication and mobile app integration				✓	✓	
System testing and optimization					✓	✓
Final documentation and report preparation						✓

10 Result

The TRASHBOT prototype successfully demonstrates autonomous navigation with obstacle avoidance in structured environments. It supports dual-mode operation through IoT-based manual and autonomous control. The robotic arm performs basic collection of small to medium dry waste, while the segregation system classifies materials into metallic and non-metallic categories using inductive sensing. The system also enables real-time video streaming and remote status monitoring. Overall, the prototype shows reliable performance in controlled indoor and flat terrain conditions.

11 Limitations

1. The current prototype is designed primarily for flat and structured environments.

2. Waste detection and collection are limited to small and medium-sized objects.
3. Battery capacity restricts the continuous operating time of the robot.
4. Obstacle detection accuracy may decrease under complex outdoor conditions.
5. The waste segregation system can only distinguish between metallic and non-metallic materials.
6. Performance may be affected by poor network connectivity during IoT communication.

12 Comparison Table

Table 4: Comparison with Related Works

System	Primary Function	Locomotion & Control	Key Innovation
TrashBot (Ours)	Auto waste collection & metal/non-metal segregation.	4-wheel drive, Raspberry Pi 3, Ultrasonic, IoT.	Dual-mode (auto/manual), robotic arm, inductive sensor, low-cost prototype.
The V Robot [2]	Indoor collection & real-time visual sorting.	Wheeled, Raspberry Pi 4, Webcam.	OpenCV visual identification, 3-joint servo grasping arm.
DustCart [5]	On-demand door-to-door urban waste collection.	Wheeled balancing, GPS, LiDAR, Ultrasonic.	SMS/call summoning, touchscreen interface, embedded air quality sensors.
Trombia Free [3]	Heavy-duty outdoor industrial street sweeping.	Fully autonomous, all-electric, 3D LiDAR, GNSS.	Filterless cyclone dust control, automated continuous emptying station.
SmartBin Rover [1]	Patrolling mobile dustbin for public areas.	Wheeled, GPS, Ultrasonic, YOLO.	AI person-tracking automatically approaches users to encourage disposal.

13 Future Plan

While core YOLO-based detection and navigation are operational, future R&D will focus on scalability, energy efficiency, and ecosystem integration. To achieve **advanced navigation**, the system will upgrade from ultrasonic sensors to high-resolution Lidar and

implement SLAM for precise mapping in complex, dynamic environments. For **smart city integration**, we plan to develop a centralized municipal dashboard for waste "heat maps" and transition to **5G / LoRa WAN** for long-range, local-hotspot-free telemetry. Additionally, enhancements in **power and autonomy** will include integrating flexible solar panels on the chassis and developing autonomous docking stations to enable uninterrupted 24/7 operation. Finally, to introduce **swarm robotics** capabilities, we aim to establish a mesh network for fleet coordination, allowing multiple units to communicate and collaborate efficiently in areas with heavy waste accumulation.

14 Conclusion

The TRASHBOT project has successfully showcased the viability of an IoT-based, autonomous waste management system aimed at tackling urban sanitation issues. The combination of the computational capabilities of the Raspberry Pi 3 Model B and the mechanical capabilities of the actuators along with the sensor fusion ensures that the system is able to fill the gap between manual labour and smart automation.

The conclusion of this prototype is the validity and feasible cost of automated hazardous waste collection. Dual-mode operation allows the flexibility in dynamic public spaces; the intelligent segregation mechanism encourages sustainable recycling processes right at the source. In short, TRASHBOT is a solid groundwork for smarter, cleaner cities to provide an upcoming waste management blueprint with great scalability.

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